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RealFact: OpenMP-Accelerated News Verification



IT 301 - Parallel Computing

National Institute of Technology Karnataka Surathkal

Introduction to the Chosen Problem

- Misinformation spreads faster than it can be checked, and style-based fake-news classifiers fail when misleading content is written in a credible tone.
- Evidence-based verification instead retrieves related articles and reasons over them rather than judging text alone.
- Comparing a claim against thousands of live articles in real time requires large-scale similarity search, which is computationally expensive.
- The task involves retrieving, classifying, and aggregating evidence from many articles per claim.

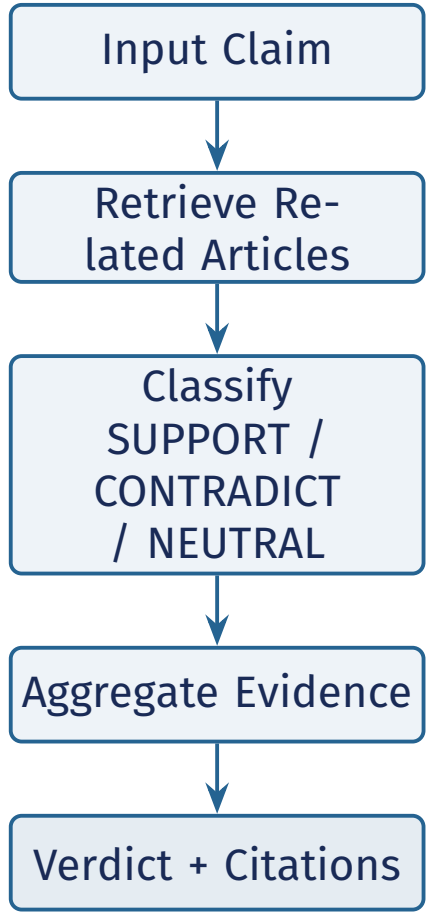


Figure : Claim-verification pipeline.

Background and Motivation

- Prior work (LIAR, FEVER, HoVer, IMRRF) advanced evidence retrieval and multi-hop reasoning for fact verification.
- These systems are almost always benchmarked on accuracy alone; latency and scalability on multicore hardware are rarely discussed.
- Syndicated or duplicate reports risk being miscounted as independent confirmations if not filtered.
- Real-time verification against a constantly updated pool of live articles needs efficient multicore parallel retrieval.

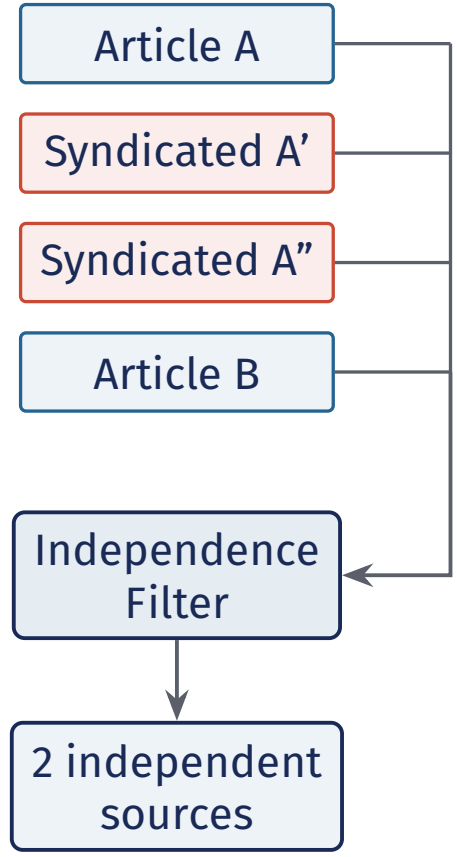


Figure : Duplicates collapsed to independent evidence.

Uniqueness/Novelty of our approach

- An evidence-independence-aware aggregation stage discounts syndicated duplicates before scoring a claim.
- An OpenMP-accelerated retrieval engine is benchmarked across static, dynamic, and guided scheduling strategies at varying thread counts.
- The system operates over live, permitted news feeds rather than a fixed offline corpus.
- Combines semantic retrieval (BGE-small-en-v1.5 embeddings) with an NLI model classifying each source as SUPPORT, CONTRADICT, or NEUTRAL.

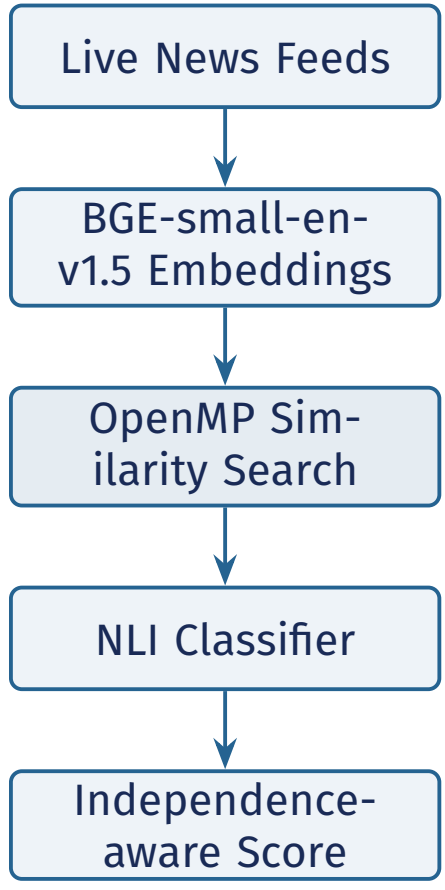


Figure : Retrieval + NLI scoring stack.

Evaluation

- Verification quality assessed on FEVER, LIAR, and FakeNewsNet using accuracy, precision, recall, F1-score, and evidence-retrieval Recall@K.
- Parallel performance measured serially and under OpenMP's static, dynamic, and guided scheduling modes.
- Tested at 1, 2, 4, 8, and 16 threads.
- Metrics: speedup, parallel efficiency, throughput, CPU utilization, and scalability against corpus size.

$$S_p = \frac{T_1}{T_p}$$

$$E_p = \frac{S_p}{p}$$

Figure : Speedup and parallel efficiency.

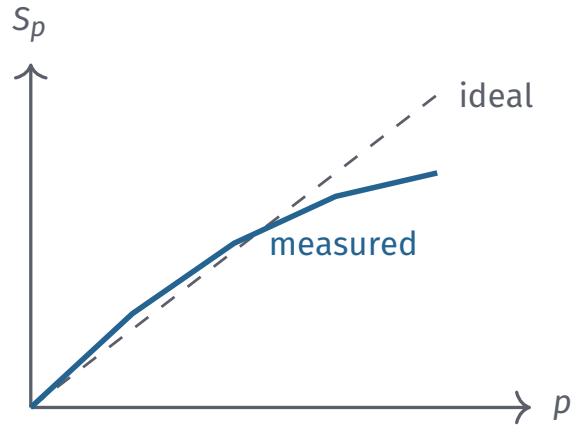


Figure : Expected speedup trend.

Expected Outcomes

1. A working prototype for real-time, multi-source news verification.
2. A C++/OpenMP semantic-retrieval engine.
3. A duplicate- and syndication-aware evidence-aggregation mechanism.
4. A comparative study of OpenMP scheduling strategies applied to this workload.
5. Explainable, evidence-cited verdicts in place of a single binary fake/real label.

Thank You

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